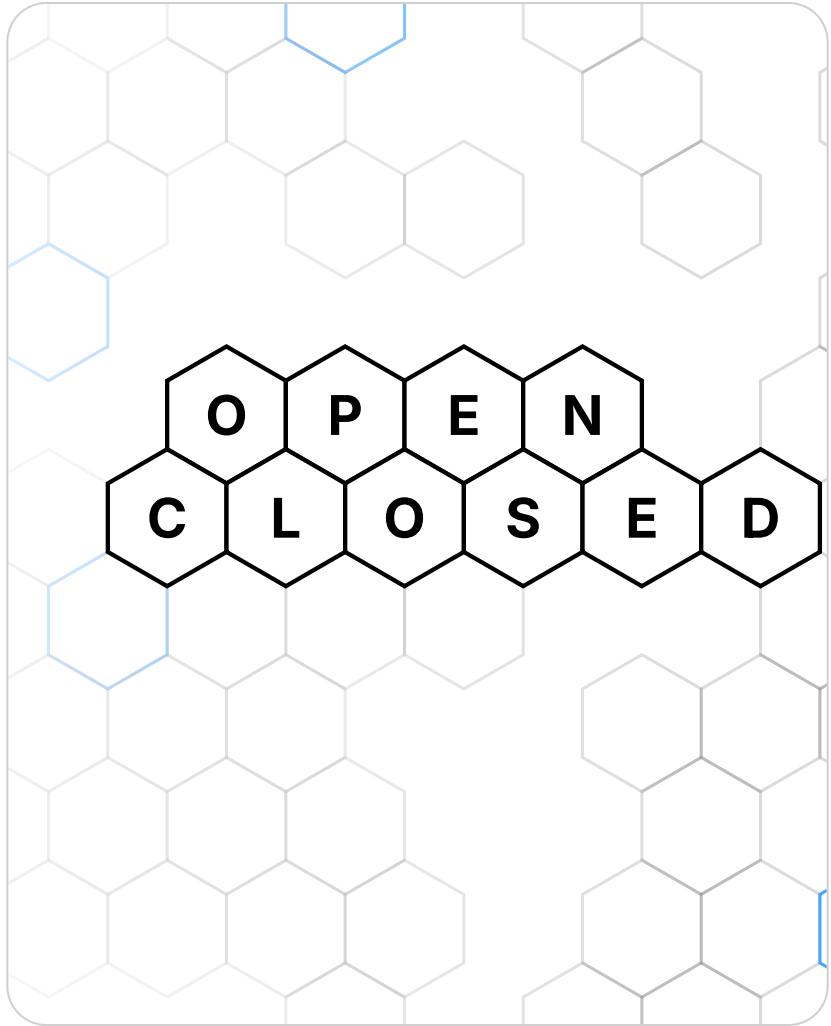
A!

OPEN SCIENCE · PERSONAL DATA · AI

As open as possible, as closed as necessary … in the age of AI

Open science with personal data plus the opportunities and risks that AI can bring: what researchers and support personnel can do



Enrico Glerean, Staff Scientist, Aalto University, November 2026

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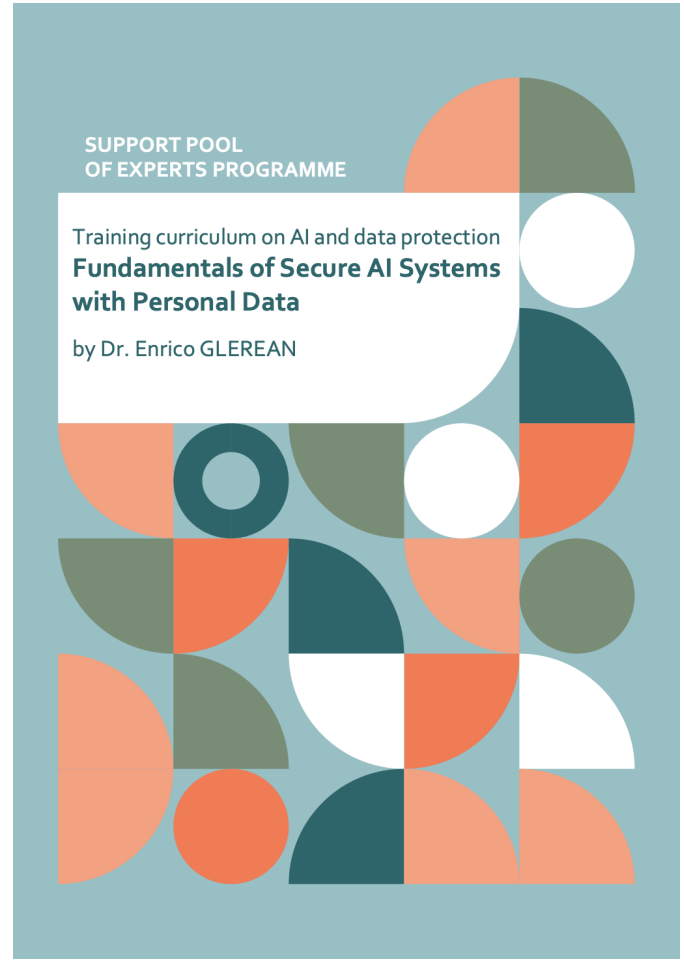
ABOUT ME

Enrico Glerean, DSc.

Staff scientist and Data Agent at Aalto University (background in neuroimaging). I train and support researchers with **handling personal data** (anonymization, secure computing), medical images, clinical trials, **research ethics and research integrity** (AI and new technologies), statistics, and **open science**. I coordinate **LUMI AI Factory trainings** at Aalto University.

Other affiliations / COI

- CodeRefinery: Nordic network to teach computational reproducibility
- Finnish Reproducibility Network: national network to raise awareness of reproducibility
- Support Pool of Experts at the European Data Protection Board: open training materials on personal data, AI and cybersecurity
- Member of the EU AI Act Advisory Forum
- Teacher in the Data Steward training programme, Tampere University

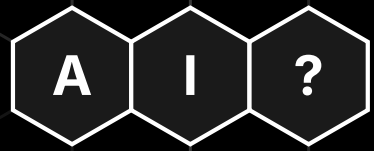


References

1. Book: Fundamentals of Secure AI Systems with Personal Data (EDPB, 2025)
2. Slides: Generative AI tools and research integrity
3. Slides: Responsible AI in research: ethics, integrity, compliance (Nordic Data Stewardship Network, 2026)
4. Slides: Data available on unreasonable request (Zenodo, 2026)
5. Slides: Talk at the Future of Privacy Forum

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What is AI?



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What is AI in research? Topic, method, tool

Reference: Best practices for AI use in research work (Aalto.fi)

AI as a topic

AI is what is being studied.

- A new ML model or architecture
- A new algorithm for training
- How people interact with AI chatbots

AI as a method

AI is how the data are analysed.

- A deep neural network that classifies tissue in cancer imaging data
- A scikit-learn model in the analysis of a dataset

AI as a tool

AI is something that helps with the work.

- Brainstorming with a chatbot
- A coding agent writing analysis scripts
- An AI assistant helping to fill in a data management plan

Invisible AI: in word processors, search engines, translation, transcription tools, automatic meeting note-takers.

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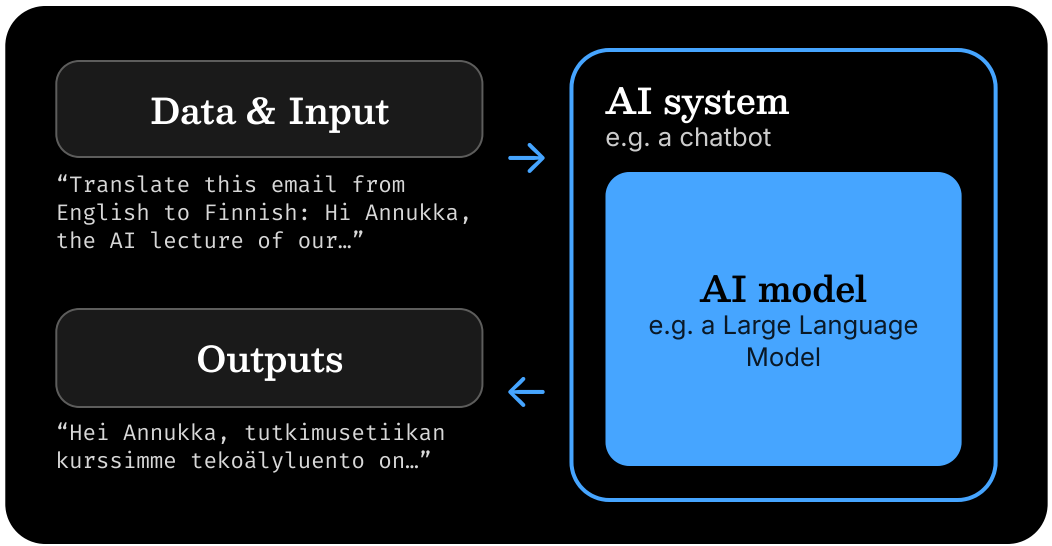
Generative AI and Research Integrity

Figure adapted from Glerean (2025) "Fundamentals of secure AI systems with personal data", European Data Protection Board

For simplicity we focus on **generative AI systems** that use **General Purpose AI models** such as **Large Language Models** that can be used in your research work, including **agentic AI** systems.

Generating one modality from another

- Text → Text
- Text → Code
- Speech → Text
- Text → Speech
- Text → Image
- Image → Text
- ...



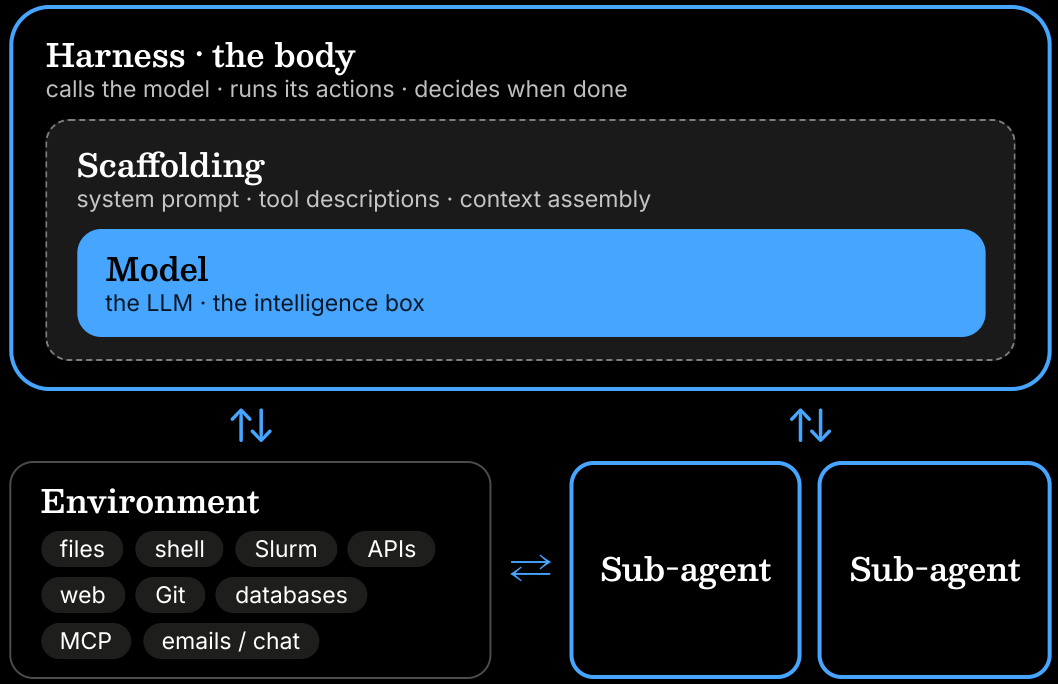
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What is an AI agent?

The four elements of an AI agent: **a) goal complexity, b) environmental complexity, c) adaptability, d) independent execution.** (OECD 2026).

An AI agent pairs the model with more components so that it can **take real actions**.

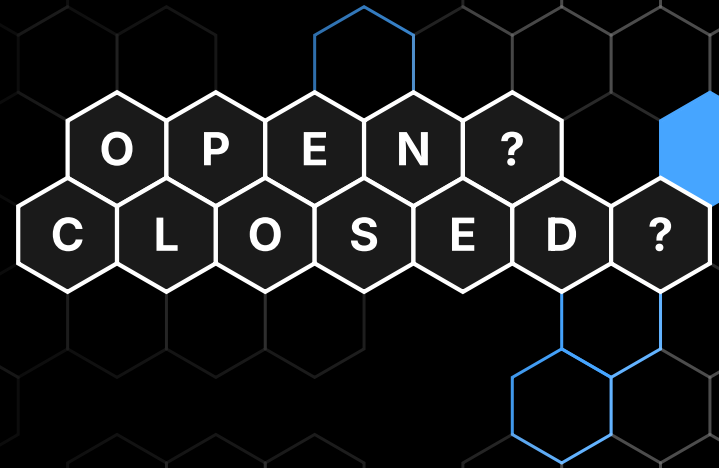
- **The model**, the "intelligence box": the LLM itself.
- **The scaffolding**, between the two: system prompts, tool descriptions, and the rules for how context is assembled and formatted before it reaches the model.
- **The harness**, the "body": the code around the model that calls it, carries out whatever it asks for, and decides when the task is done.



Adapted from Hugging Face, "Harness, Scaffold, and the AI Agent Terms Worth Getting Right", 2026

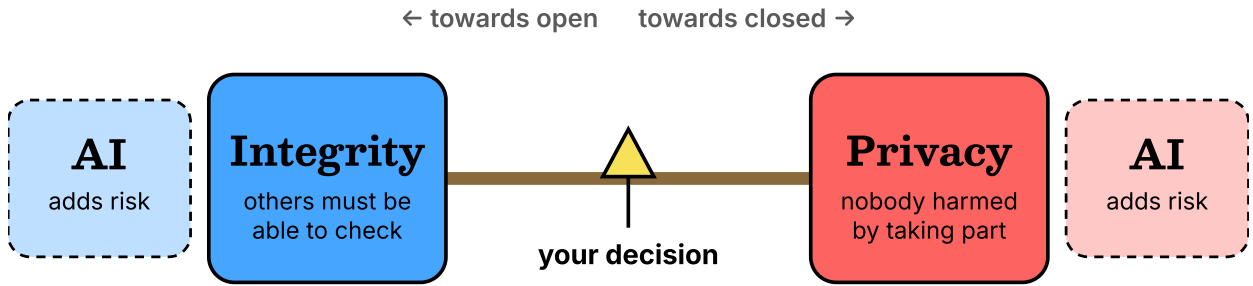
The tension

Open science versus data privacy



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One tension, many added risks



Integrity: the ALLEA principles

- **Reliability** in the design, method, analysis and use of resources
- **Honesty** in developing, reviewing, reporting and communicating research
- **Respect** for colleagues, participants, research subjects, society, ecosystems and the environment
- **Accountability** for research from idea to publication

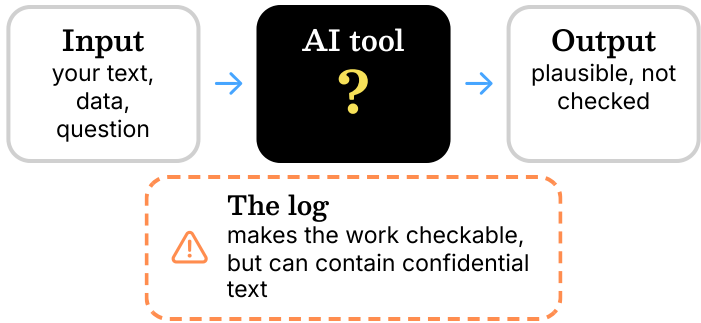
Privacy: the 7 principles of privacy by design

1. Proactive, not reactive
2. Privacy as the default setting
3. Privacy embedded into design
4. Full functionality: positive-sum, not zero-sum
5. End-to-end security across the data life cycle
6. Visibility and transparency
7. Respect for user privacy

ALLEA 2023 · Cavoukian 2009 · GDPR Art. 25: data protection by design and by default

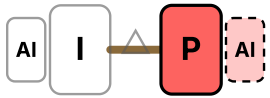
Generative AI: the epistemic risks

- **The ideal misconduct machine?** Generative AI always produces text and leaves quality control to the user. Unchecked output can turn into **fabrication, falsification or plagiarism**, and **hiding AI use is itself an integrity violation**, listed next to them (ALLEA 2023 §3.1).
- **Errors get cheaper.** Papers citing references that do not exist: 1 in 2,828 in 2023, 1 in 277 in early 2026.
- **The same prompt can give different outputs**, so "we used ChatGPT" does not let anyone reconstruct the work.
- **Peer review is exposed too:** authors have hidden instructions in manuscripts to steer AI reviewers.
- **The catch:** prompts and logs make AI use checkable, but they can contain confidential text. That is the tension again.



Topaz 2026 · EC Guidelines v4 · Cui & Alexander 2026 · ALLEA 2023 · Nikkei 2025 · More cases: AI tools, Research Integrity and Ethics

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Generative AI: the governance risks

- **Agents act, they do not just answer.** An AI agent can read files, run code and use your credentials. Whatever it can reach, anyone who takes control of it can reach too.
- **Hidden instructions hijack agents.** Text in a web page, email or document can tell an agent to send data elsewhere (prompt injection). There is no reliable fix yet.
- **Attacks get cheaper.** In 2025 one actor used an AI coding agent to steal data from at least 17 organizations in about a month, healthcare among them.
- **Re-identification gets cheaper.** 6 of 24 public interview transcripts were linked to publications, at under $0.50 each (preprint: Li 2026).
- **Sovereignty is a security control.** A US provider can be compelled to hand over data even from EU data centres (CLOUD Act). **Sovereign AI**, open models on European infrastructure such as the LUMI AI Factory, keeps data under EU law.

Closed data, breached

ATTACK Vastaamo, 2020: therapy records stolen, patients blackmailed.

ATTACK University of Manchester, 2023: data on 1.1 million NHS patients.

MISTAKE Microsoft AI research, 2023: a model-sharing link exposed 38 TB.

AI AGENT Data extortion, 2025: an AI coding agent broke in, stole data and wrote the ransom notes.

MISUSE UK Biobank, 2026: approved users broke the rules.

Not only personal data

Drafts, code and ideas sent to remote AI tools are confidential too.

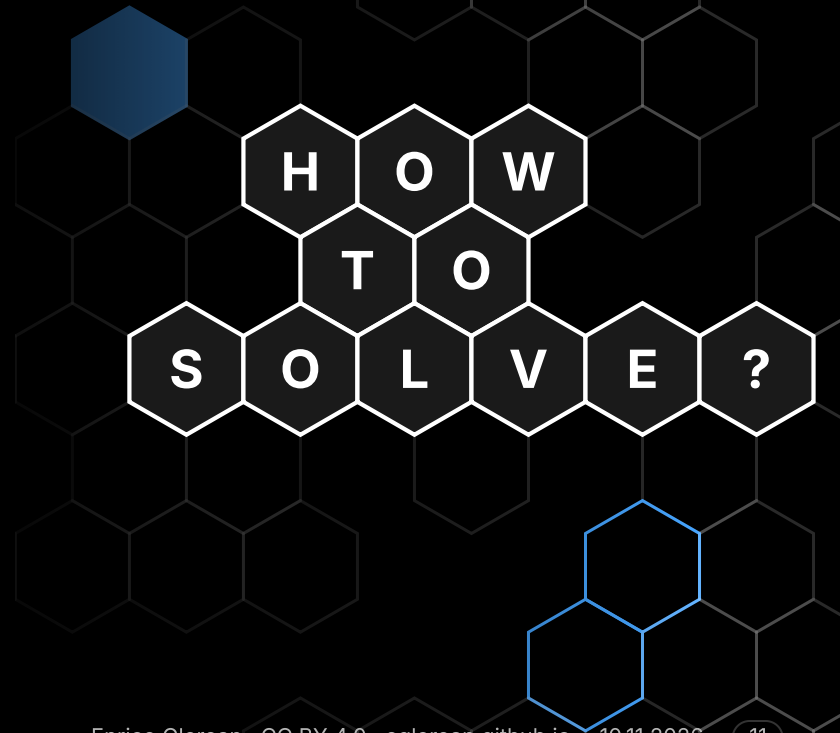
DISPUTED Navier–Stokes, 2026: mathematicians who checked unpublished work with Codex ask whether it fed OpenAI's result. OpenAI: unlikely, cannot rule it out.

DISPUTED Claude's "new" enzyme, 2026: a Copenhagen researcher who shared drafts with Claude had found the enzymes first. Anthropic: no access to his chats.

OWASP LLM01 2025 · Anthropic 2025 · Li 2026 · CLOUD Act · LUMI AI Factory · More: AI tools, Research Integrity and Ethics

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What should researchers and research support do?



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1. For each object pick a level of protection

- A study is many objects. Open or closed applies to **each one**, not to the study. For each object, record the **access level**, the **reason** and the **route** to access.
- FAIR does not mean open: access can require login and approval, while the description stays public. "On reasonable request" gives no reason and no route.

Object RUNNING CASE: 80 INTERVIEWS	Level	Reason	Route
Protocol, interview guide	**Public**		Repository with DOI
Codebook, data dictionary	**Public**	Check example quotes first	Repository
Analysis code	**Public**	Check for embedded data first	Repository
Summary tables	**Public**	Merge very small groups	Article
AI workflow description	**Public**		Supplement
AI prompts and logs	**Controlled**	Contain interview text	Audit request
Transcripts	**Controlled**	Participants could be recognised	Data access committee, agreement, secure environment
Key linking codes to names	**Not shared**	Identifies everyone	None

ALLEA 2023 (§2.5, §3.1) · Open Data Directive · Horizon Europe · FI policy 2026–2030 · Wilkinson 2016 · van Ravenzwaaij 2025

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2. PETs and who could identify whom

- Chose the right combination of **Privacy Enhancing Technologies** (PETs) to minimize personal data
- **Pseudonymised data is still personal data.**
- **The same data can be personal for one party and not for another** (*EDPS v SRB*). Public release is the hard case: anyone, with any tool.
- **Is it anonymous? three checks:** can someone be singled out, linked to other data, or have something inferred about them? (see EDPB guidelines draft)
- **Promise only what you can keep.** Do not write "anonymous" for a small interview study, and recheck as tools improve.

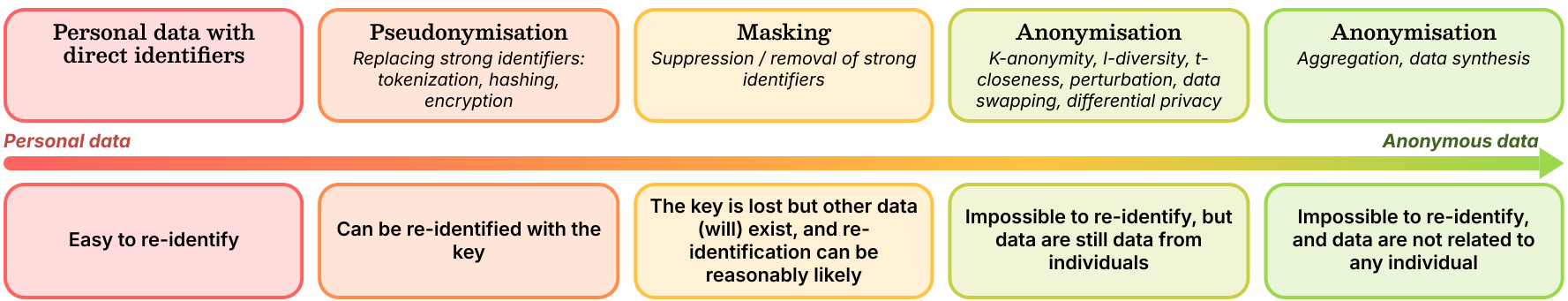


Figure: Glerean 2025, Fundamentals of Secure AI Systems with Personal Data (EDPB) · EDPB 02/2026 (draft) · Knolle 2026

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3. Match the tool to the data



- **Ask where it runs, not which brand:** who receives the data, where they are stored, how long they are kept, whether they train the model, who can look.
- **Keep the full AI log restricted.** Publish a clean workflow: tool and version, task, input type, what you accepted, who checked.
- **Never give one AI agent** private data, untrusted content and a way to send data out. Remove one, for example the internet connection.

Public	→	Any tool
Internal	→	Approved tools
Confidential	→	Contractual guarantees or local models
Secret	→	Local tools inside a secure environment

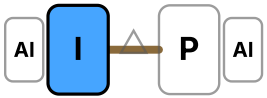
Full AI log *restricted* → FILTER → **Public workflow** *with the paper*

RUNNING CASE Pseudonymised transcripts do not go into a consumer chatbot.

Aalto best practices · EC Guidelines v4 · TENK 2026 · Cwik 2026 · Willison 2025

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4. Keep closed work checkable





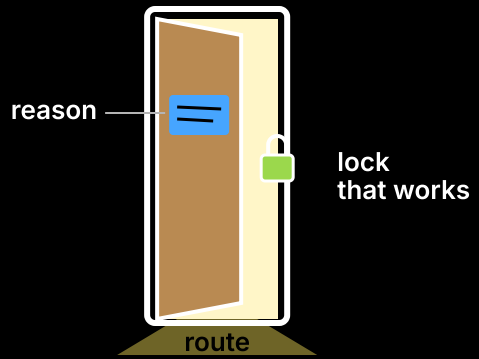
↳ second path: a **qualified verifier** reruns the same code on the same data version inside the secure environment

- **Sharing code is not enough.** One team ran the same scripts unchanged in a second secure environment and still hit problems with how the data had been prepared.
- **Publish what you can:** code, data dictionary, data version, software setup, access route.
- **Say who can verify the work and how**, for example a qualified person rerunning it inside the secure environment.
- **Plan for time.** Controlled clinical trial data took 270 days on average to access, open data 8.
- **This is the EU direction.** From 2029, under the European Health Data Space: permit, secure environment, only non-personal outputs leave, results public within 18 months.

Williams 2025 · Rodriguez 2025 · EHDS (Art 61) · Findata

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Four questions before you open, close or upload



1. Which object is this, and why is it closed?
2. Who could identify whom, with what means today?
3. Where does it run, and who else can see it?
4. How can someone still check it?

AI use: generative AI tools were used to code the deck as HTML pages and retrieve some sources.

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Privacy Enhancing Technologies (PETs)

Broadly two families:

- Techniques that **transform the data**
- Techniques that **harden the computing environment**

Embedded in the security of all other components.

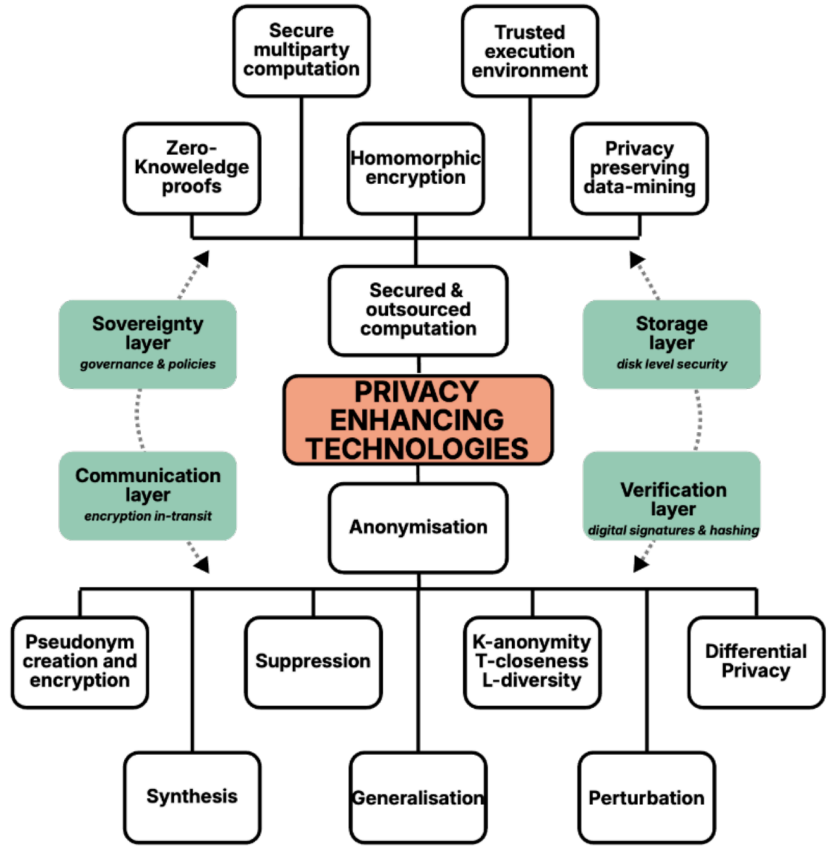


Figure 5.1 from Glerean 2025. The landscape of PETs solutions. Various techniques can be combined to ensure the privacy of the entire system by altering the data or by improving security of storage, communication, and computations. The image is an evolution of Garrido et al. (2022).

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BACKUP

Techniques move risk, they do not remove it

Technique	Gives you		Risk moves to
Pseudonymisation	No names in working files	→	Anyone who can link; stories still identify
Synthetic data	Teaching, software tests, previews	→	Lost patterns, leftover disclosure risk
Differential privacy	A formal limit on one person's influence	→	Accuracy, especially for small groups
Federated learning	Raw data stay at each site	→	Updates can leak; site differences cost accuracy
Secure environment	Control of who, where and what leaves	→	Cost, usability, reproducibility

Adams 2025 · Demuth 2025 · Federated learning review 2025 · Honkela interview 2026

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BACKUP

Check the status before you cite the rule

Binding now	Phasing in	Not law DRAFT PROPOSAL
• GDPR • Open Data Directive, through national law • AI Act: prohibitions, GPAI, Art 50 • Digital Omnibus on AI, Reg. (EU) 2026/1744, in force 27 Jul 2026 • Finnish Secondary Use Act	• CRA: reporting since 11 Sep 2026, full 11 Dec 2027 • AI Act high-risk: Annex III 2 Dec 2027, Annex I 2 Aug 2028 • EHDS: 26 Mar 2027; secondary use 26 Mar 2029; more categories 26 Mar 2031	• EDPB draft Guidelines 1/2026 (research) • EDPB draft Guidelines 02/2026 (anonymisation) • Digital Omnibus COM(2025)837: Council mandate 21 Oct 2026, Parliament committee vote expected early 2027

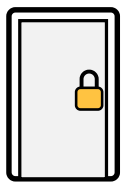
Scope trap: the AI Act excludes research-only systems and pre-market research (Art 2(6), 2(8)). Deployment and real-world testing are not excluded.

AI Act · AI Omnibus · EHDS · CRA · EDPB 1/2026 · EDPB 02/2026 · COM(2025)837 · Council mandate · NicFab 2026

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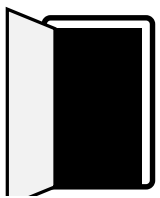
THE TENSION

Three ways to fail the same principle



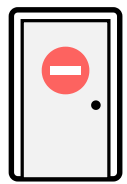
Closed by habit

"Available on request": of 1,792 authors asked, 123 shared their data (6.8%).



Open by accident

UK Biobank data ended up in public code repositories, inside code that journals often ask authors to publish.



Closed on paper

April 2026: data on about 500,000 volunteers listed for sale. No hack: approved institutions broke the rules.

Gabelica 2022 · UK Biobank statement 2026 · UKB Oversight Committee 2026 · The Register 2026

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